

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

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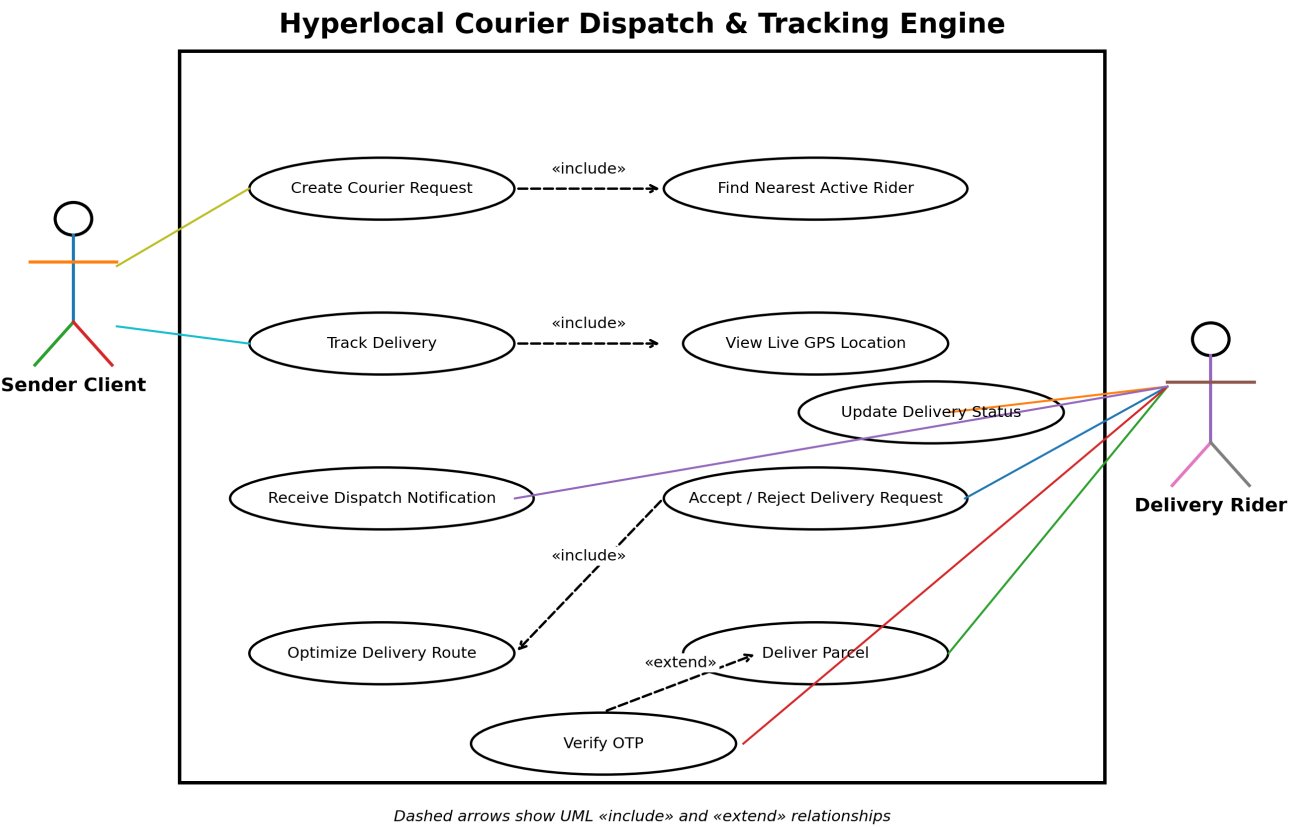
Problem Statement #23: Hyperlocal Courier Dispatch & Tracking Engine

Deliverable 1: Complete Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: Rider within 3 km receives dispatch notification. Fail: Offline rider is assigned.	Fast and efficient rider assignment.
FR-002	Functional	Allow the sender to create a courier request with pickup, destination, parcel, and recipient details.	High	Pass: Request is created with all mandatory data. Fail: Request is created with missing data.	Required to initiate delivery.
FR-003	Functional	Allow the rider to accept or reject an assigned courier request.	High	Pass: Rider response updates status. Fail: Status remains pending after response.	Confirms rider availability.
FR-004	Functional	Provide real-time delivery status and GPS location updates to the sender.	High	Pass: Sender views current status and latest location. Fail: Updates unavailable during active delivery.	Provides transparency and tracking.
FR-005	Functional	Require OTP verification before marking the courier as successfully delivered.	High	Pass: Delivery completes only after correct OTP. Fail: Delivery completes without OTP.	Improves delivery security.
NFR-001	Performance & Security	Transmit delivery status and live GPS telemetry with under 2-second latency.	High	Pass: Tests confirm under 2-second latency under simulated peak load. Fail: Target is consistently exceeded.	Supports real-time tracking.
NFR-002	Reliability & Availability	Maintain accurate courier request and delivery status information during normal operations.	High	Pass: Data remains accessible and consistent. Fail: Requests are lost, duplicated, or incorrect.	Reliable information is essential for dispatch and tracking.

Deliverable 2: UML Use-Case Diagram

Actors: Sender Client and Delivery Rider. The diagram shows the major system interactions and UML relationships clearly.



Deliverable 3: Use-Case Flow Specification

UC-001: Create and Dispatch Courier Request

Primary Actor: Sender Client

Supporting Actor: Delivery Rider

Preconditions

- The Sender Client has access to the courier platform.
- The required pickup, destination, parcel, and recipient details are available.
- The delivery service is operational in the requested area.

Postconditions

Success: A courier request is created, a suitable rider is assigned, and the sender is notified.

Failure: The request remains unassigned and the sender is informed if no suitable rider is available.

Main Success Scenario

1. The Sender Client selects the option to create a new courier request.
2. The system requests pickup location, destination location, parcel details, and recipient information.
3. The Sender Client enters the required information.
4. The system validates the submitted information.
5. The system creates the courier request.
6. The system searches for the nearest active delivery rider within a 3 km radius.
7. The system selects a suitable available rider.
8. The system sends a dispatch notification to the Delivery Rider.
9. The Delivery Rider receives the dispatch notification.
10. The Delivery Rider accepts the delivery request.
11. The system assigns the courier request to the rider.
12. The system updates the courier status to Assigned.
13. The Sender Client is notified that a delivery rider has been assigned.

Alternate Flow: No Rider Available

1. The system searches for an active rider within the 3 km radius.
2. No suitable active rider is found.
3. The system does not assign the request to an offline or unavailable rider.
4. The courier request remains pending.
5. The Sender Client is informed that no rider is currently available.